AIRLINE LOYALTY PROGRAM CHURN ANALYSIS AND RETENTION STRATEGY

Executive Summary

Airline loyalty programs are designed to increase customer retention and encourage repeat travel. However, customers often begin disengaging long before they formally leave a loyalty program. The objective of this project was to identify customers at risk of disengagement, understand the behavioral drivers of churn, develop meaningful customer segments, and recommend practical retention strategies.

The analysis revealed that customer behavior is a significantly stronger predictor of churn than traditional value-based measures such as Customer Lifetime Value (CLV). Approximately 15.7% of customers exhibited behavioral churn, while a predictive model achieved 94.5% accuracy in identifying future churn risk. Most importantly, a group of 660 "Silent High Value" customers was identified. These customers possess exceptionally high historical value but currently exhibit low engagement and elevated churn risk, representing approximately $11.4 million in customer value at risk.

The findings suggest that retention efforts should shift from value-based targeting toward behavior-based intervention strategies.

1. Problem Framing

1.1 Business Context

Customer retention is one of the most important objectives of any airline loyalty program. Retaining an existing customer is generally less expensive than acquiring a new one, making loyalty programs a critical component of long-term profitability.

While airlines typically focus on customer value metrics such as CLV, these measures often fail to identify customers who are gradually disengaging from the program. As a result, high-value customers may become inactive without receiving any targeted intervention.

1.2 Business Problem

The airline faces three key challenges:

1. Identifying customers who are beginning to disengage.
2. Understanding the factors that contribute to churn.
3. Prioritizing retention resources toward customers who matter most.

The central question addressed in this project was:

"Which customers are most likely to disengage from the loyalty program, and what actions should the airline take to retain them?"

1.3 Project Objectives

The objectives of this analysis were:

- Measure customer-level behavioral churn.
- Identify the key drivers of customer disengagement.
- Develop meaningful customer segments.

- Predict future churn risk.
- Recommend targeted retention strategies.

2. Data Preparation and Analytical Approach

2.1 Dataset Overview

The dataset contained airline loyalty program information including customer activity, loyalty membership details, reward redemption behavior, and Customer Lifetime Value (CLV).

Key variables included:

- Total Flights
- Distance Flown
- Points Accumulated
- Points Redeemed
- Loyalty Card Tier
- Enrolment Year
- Customer Lifetime Value (CLV)

2.2 Data Cleaning Decisions

Several preprocessing decisions were required before analysis.

First, monthly activity records were aggregated into customer-level profiles. This was necessary because the objective was to evaluate overall customer behavior rather than individual monthly transactions.

Second, duplicate activity records were removed to ensure accurate calculations of customer engagement.

Third, customer activity was summarized using unique month-level participation rather than raw row counts. This prevented customers with duplicate records from appearing artificially more active.

These cleaning decisions ensured that engagement metrics reflected actual customer behavior rather than data structure artifacts.

2.3 Feature Engineering

Several new variables were created to support analysis.

- Behavioral Churn: A customer was classified as behaviorally churned if they experienced six consecutive months of inactivity. This definition was selected because it captures disengagement before formal customer loss occurs.
- Active Months: The number of months in which a customer recorded flight activity.
- Redemption Ratio: The proportion of accumulated points that were redeemed by the customer.
- Engagement Score: A composite metric designed to measure customer engagement using activity consistency, flight frequency, and reward participation.

These variables were specifically created to evaluate customer behavior rather than historical customer value.

3. Exploratory Analysis and Key Findings

3.1 Overall Behavioral Churn

Customer-level analysis revealed a behavioral churn rate of approximately 15.7%.

This indicates that nearly one in six customers experienced a prolonged period of inactivity during the observation period.

Figure 1: Executive Overview Dashboard showing overall customer activity, churn rate, loyalty card distribution, enrolment trends, and engagement metrics.

3.2 Loyalty Tier Does Not Explain Churn

An initial assumption was that customers in higher loyalty tiers would exhibit substantially lower churn rates.

However, behavioral churn rates were remarkably similar across loyalty card tiers, ranging from approximately 14.9% to 17.0%.

This suggests that loyalty tier alone is not a reliable indicator of future customer retention.

3.3 Reward Redemption Supports Retention

One of the strongest findings involved reward redemption behavior.

Customers who redeemed loyalty points exhibited significantly lower churn rates than customers who never redeemed rewards.

This suggests that reward redemption functions as an engagement mechanism rather than simply a program benefit.

3.4 CLV Alone Does Not Explain Churn

Another surprising finding was that churn rates remained relatively consistent across customer value segments.

Historically valuable customers were not automatically more loyal than lower-value customers.

This challenged the assumption that CLV alone should guide retention strategy.

3.5 Most Interesting Finding: Silent High Value Customers

The most important discovery was the identification of a segment of customers who had generated substantial historical value but were currently exhibiting low engagement and high churn risk.

These customers were classified as "Silent High Value" customers.

This group emerged as the airline's highest-priority retention opportunity.

4. Understanding Different Customer Types

4.1 Segmentation Logic

Rather than applying unsupervised clustering techniques directly, customer segments were designed around behavioral patterns identified during exploratory analysis.

This approach was chosen because it produces segments that are easy to interpret and directly linked to business actions.

4.2 Identified Customer Segments

Six customer segments were created:

- Silent High Value: High-value customers exhibiting low engagement and elevated churn risk.
- Reward-Oriented Active: Customers who actively engage with the loyalty ecosystem through reward redemption.
- Engaged Loyalists: Customers demonstrating consistently high engagement and activity levels.
- Regular Customers: Customers exhibiting moderate but stable participation.
- New Promotion Users: Recently acquired customers enrolled through promotional campaigns.
- Dormant Low Engagement: Customers with very low activity and minimal engagement.

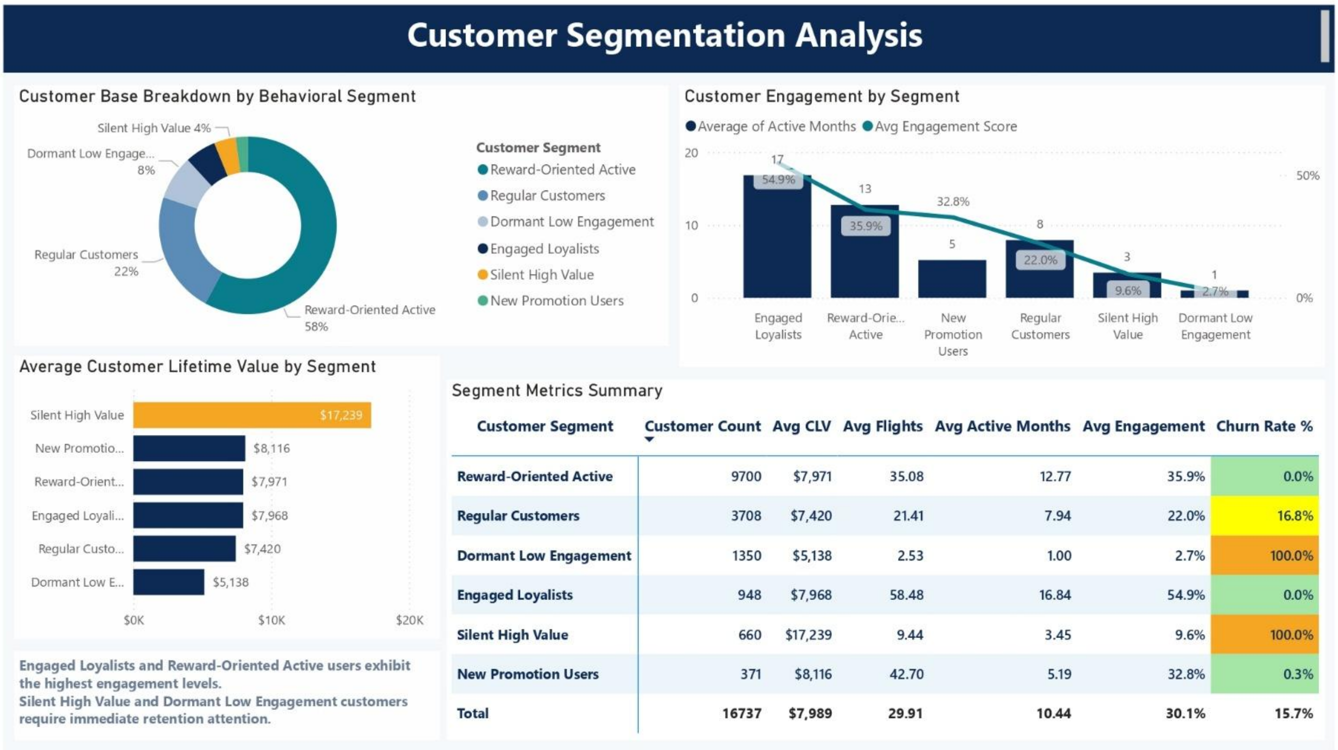


Figure 2: Customer Segmentation Dashboard highlighting customer groups, engagement patterns, and churn characteristics.

4.3 Segment Insights

The Silent High Value segment consisted of 660 customers with an average CLV of approximately $17,239.

Despite their historical value, these customers demonstrated low engagement and high churn risk, making them the highest-priority segment for retention efforts.

5. Predicting Future Customer Churn

5.1 Modeling Approach

The objective of predictive modeling was to identify customers likely to experience future behavioral churn.

A Random Forest model was selected because it effectively captures complex relationships between customer activity, loyalty behavior, and engagement metrics.

The focus was not simply achieving high accuracy, but identifying actionable drivers of churn.

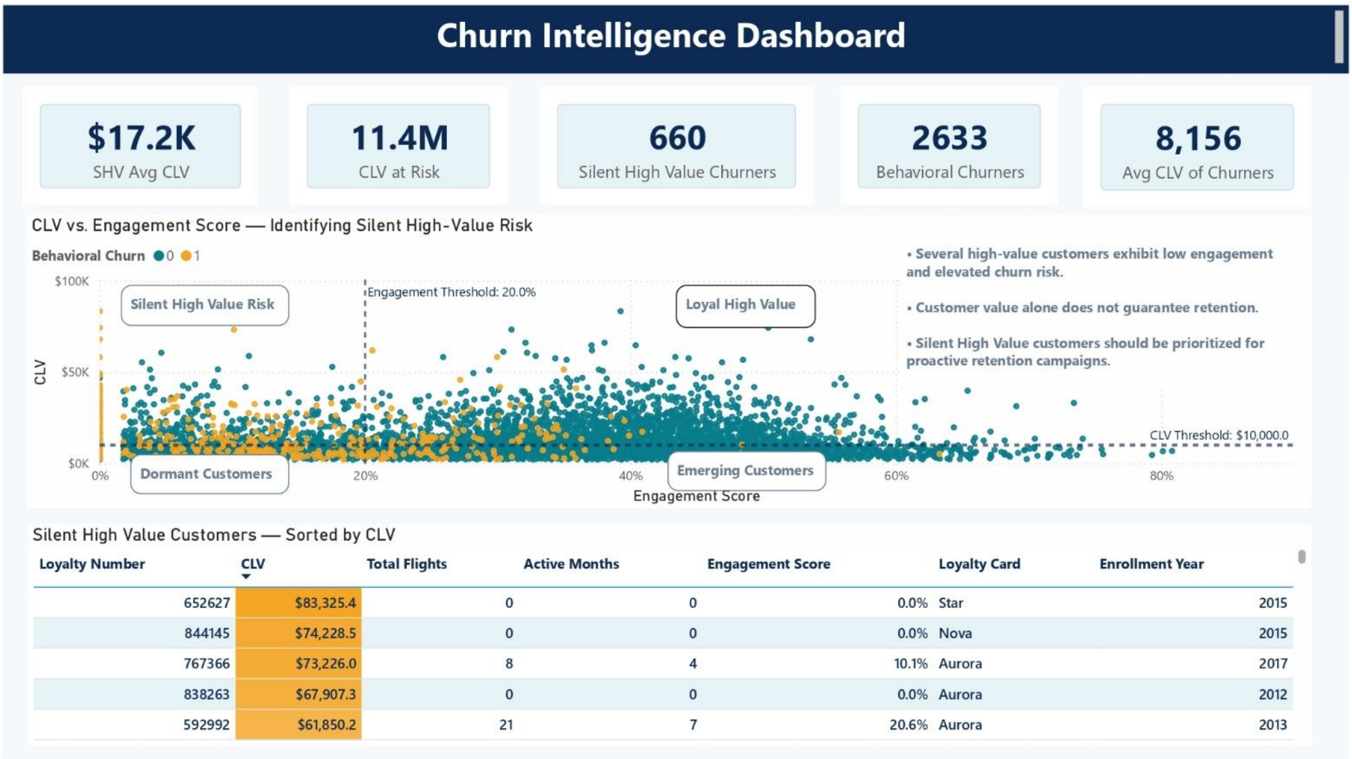
5.2 Model Performance

The final model achieved:

- Accuracy: 94.5%
- Precision: 89.6%
- Recall: 68.4%
- F1 Score: 77.6%

These results indicate strong predictive performance while maintaining a relatively low false-positive rate.

5.3 What Drives Churn?



Loyalty Number	CLV	Total Flights	Active Months	Engagement Score	Loyalty Card	Enrollment Year
652627	$83,325.4	0	0	0.0%	Star	2015
844145	$74,228.5	0	0	0.0%	Nova	2015
767366	$73,226.0	8	4	10.1%	Aurora	2017
838263	$67,907.3	0	0	0.0%	Aurora	2012
592992	$61,850.2	21	7	20.6%	Aurora	2013

Figure 3: Predictive Churn Dashboard showing model performance, confusion matrix, and feature importance analysis.

Feature importance analysis revealed three dominant predictors:

1. Points Accumulated (23%)
2. Distance Flown (20%)
3. Customer Tenure (18%)

Behavioral variables consistently outperformed traditional value-based metrics.

Notably, CLV ranked only fifth among predictive factors.

This reinforces one of the central conclusions of the project: customer behavior is a stronger predictor of future churn than historical customer value.

6. Business Recommendations

The findings support three recommendations that align with both financial and marketing objectives.

6.1 Recommendation 1: Prioritize Silent High Value Customers

The airline should immediately target Silent High Value customers through premium win-back campaigns, personalized route-based offers, and bonus-mile incentives.

These customers represent the largest concentration of value at risk.

6.2 Recommendation 2: Increase Reward Redemption Participation

Reward redemption was strongly associated with retention.

Marketing efforts should encourage first-time redemption through targeted campaigns and redemption incentives.

Increasing redemption participation is likely to improve long-term engagement.

6.3 Recommendation 3: Shift from Value-Based to Behavior-Based Retention

Retention programs should prioritize behavioral indicators such as engagement score, active months, and redemption activity rather than relying solely on CLV.

This approach allows earlier intervention before customer disengagement becomes permanent.

7. Business Impact and Strategic Value

7.1 Customer Value at Risk

The Silent High Value segment represents:

- 660 customers
- Average CLV: $17,239
- Total CLV at Risk: approximately $11.4 million

7.2 Scenario Analysis

If the airline reduces churn among Silent High Value customers from 100% to 50%, approximately $5.7 million in historical customer value could be preserved.

While this represents an illustrative estimate rather than a guaranteed outcome, it demonstrates the potential financial impact of targeted retention initiatives.

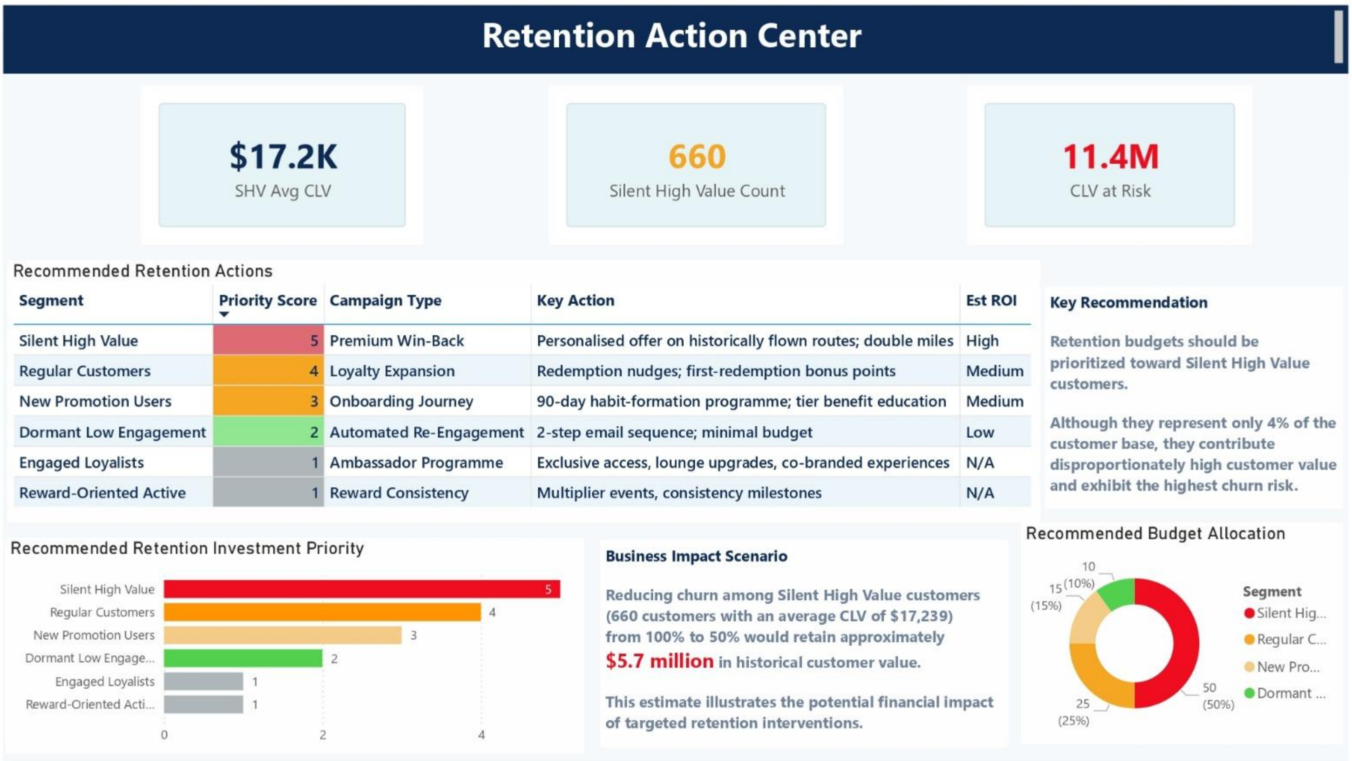


Figure 4: Retention Action Centre Dashboard showing retention priorities, campaign recommendations, and estimated business impact.

8. Conclusion

This project demonstrates that customer engagement behavior provides earlier and more reliable warning signals of churn than traditional value-based measures.

The analysis identified a meaningful behavioral churn problem, developed actionable customer segments, produced a high-performing predictive model, and translated analytical findings into practical business recommendations.

Most importantly, the findings suggest that retention decisions should be driven by behavioral engagement rather than historical value alone.

By focusing on customers who exhibit early signs of disengagement, the airline can allocate retention resources more effectively, improve loyalty program performance, and protect millions of dollars in customer value.